

EU AI Ethics Assistant — Expert Agent with Gemini, RAG and LangGraph

A conversational expert agent on **European AI ethics and regulation**, built for the Generative AI course final project.

Why this domain?

AI adoption is accelerating faster than public understanding of its risks and rules. This assistant makes ~170 pages of EU regulation, ethics guidelines and academic research accessible through natural conversation — for developers, students and policymakers who need answers, not homework.

Knowledge Base (4 sources, ~170 pages)

Document	Type	Covers
EU AI Act	Legal text	Risk categories, obligations, prohibited practices
Bias & Fairness Survey (Mehrabi et al.)	Academic paper	Bias types, fairness definitions, mitigation
EP Study: Ethics of AI	Policy study	Real-world case studies (health, justice, warfare)
Ethics Guidelines for Trustworthy AI	EU framework	7 requirements for trustworthy AI

Architecture

PDFs → cleaning → chunking (1000 chars / 150 overlap) → Gemini embeddings
(`gemini-embedding-001`) → **ChromaDB** → **LangGraph** agent
(retrieve → generate) with `gemini-2.5-flash` + **MemorySaver** conversation memory.

Installation

1. Clone the repo and `cd` into it
2. `python -m venv .venv && source .venv/bin/activate`
3. `pip install -r requirements.txt`
4. Create `.env` with `GOOGLE_API_KEY=your_key` (get one at <https://aistudio.google.com/apikey>)
5. Place the 4 PDFs in `data/` (links in `data/README` or below)
6. Open `notebooks/ai_ethics_assistant.ipynb` and run top to bottom

System Prompt Justification

(Paste the justification block from notebook Section 6 here.)

Requirements

Python 3.10+, Gemini API key. Full dependency list in [requirements.txt](#).